

DATA ANALYTICS EXERCISE 2

Q1 list all distinct departments in the student table

Expected columns: department

Table Students:

student_id	name	age	department
1	Alice	20	IT
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eva	22	HR

A1

SYNTAX

```
SELECT DISTINCT department,  
FROM students;  
TABLE
```

department
IT
HR
Finance

DATA ANALYTICS - EXERCISE 2

Q1) List all distinct department in the student table.
Expected columns: department, avg-age

TABLE students

student-id	name	age	department
1	Alice	20	IT
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eva	22	HR

Syntax
Q2) SELECT department,
AVG (age) AS avg-age
FROM students
GROUP BY department;

TABLE students

department	avg-age
IT	20.5
HR	22
Finance	23

Q2

DATA ANALYTICS EXERCISE 2

Q3 Show departments with more than 1 student.
Expected columns: department, student_count

student_id	name	age	department
1	Alice	20	IT
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eva	22	HR

Q3

Syntax

```
SELECT department
COUNT (*) AS students
FROM students
GROUP BY department
HAVING COUNT (*) > 1;
```

TABLE

TABLE students

department	student_count
IT	2
HR	2

Q4) Get all students whose age is between 21 and 23
 expected columns: student_id, name, age, department.

Student-id	name	age	department
1	Alice	20	IT
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

A4

SQL

```
SELECT student_id, name, age, department
FROM students
WHERE age >= 21 AND age <= 23;
```

OR

```
SELECT student_id, name, age, department
FROM students
WHERE age BETWEEN 21 AND 23;
```

TABLE

TABLE students

student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR
		22	Finance
			HR

DATA ANALYTICS EXERCISE 2

Q5 List all students in the IT or HR departments who are older than 21.

EXPECTED columns: student_id, name, age, department

TABLE student

student_id	name	age	department
1	Alice	20	IT
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eva	22	HR

A5

SYNTAX

SELECT *

FROM students

WHERE department IN ('IT', 'HR') AND age > 21

TABLE student

student_id	name	age	department
3	Charlie	21	IT
2	Bob	22	HR
5	Eva	22	HR

DATA ANALYTICS EXERCISE 2

- Q6 Show total credits per department, only for departments with more than 5 total credits
Expected columns: department, total credits

TABLE

course-id	course-name	department	credits
101	SQL Basics	IT	3
102	Python	IT	4
103	Data Science	IT	4
104	Excel	Finance	2
105	Statistics	HR	3

A6

SYNTAX

SELECT department, SUM(credits) AS total-credits

FROM course

GROUP BY department

HAVING SUM(credits) > 5;

TABLE

department	credits
IT	11

DATA ANALYTICS EXERCISE 2

Q4 List all courses that do not have 4 credits
expected columns : department, course id, course name, credits

Table courses

course-id	course-name	department	credit
101	SQL Basics	IT	3
102	Python	IT	4
103	Data Science	IT	4
104	Excel	Finance	2
105	Statistics	HR	3

A7

SYNTAX

SELECT *

FROM courses

WHERE credits <> 4;

TABLE

Table course

course-id	course-name	department	Credit
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

DATA ANALYTICS

(Q8) Show the top 3 course by credits in descending order
EXPECTED Column: course-id, course-name, credits

course-id	course-name	department	credits
101	SQL Basics	IT	3
102	Python	IT	4
103	Data Science	IT	4
104	Excel	Finance	2
105	Statistics	HR	3

(A8)

SYNTAX

```
SELECT course-id, course-name, credits  
FROM courses  
ORDER BY credits DESC  
LIMIT 3;
```

TABLE Course

course-id	course-name	credits
103	Data Science	4
102	Python	4
101	SQL Basics	3

DATA ANALYTICS EXERCISE 2

3 Enrolment Table

(Q9) Get the maximum and average grade across all enrolments.

EXPECTED columns: max grade min grade avg grade

TABLE Enrolments

enrolment_id	student_id	course_id	grade
1	1	101	85
2	2	102	78
3	3	103	90
4	4	104	80
5	5	105	82

(A9) SYNTAX

```
SELECT MAX (grade) AS max-grade,  
       MIN (grade) AS min-grade,  
       AVG (grade) AS avg-grade,  
FROM enrolments;
```

max-grade	min-grade	avg-grade
90	78	83

EXERCISE 2 Data Analytics (35 marks, 15 min)

Q10 Count how many enrolments exist per course
EXPECTED columns: course-id, enrolment-count,

TABLE enrolments

enrolment-id	student-id	course-id	grade
1	1	101	85
2	2	102	78
3	3	103	90
4	4	104	88
5	5	105	82

Q10 SYNTAX

```
SELECT count_id
COUNT (*) AS enrolment-count
FROM enrolments
GROUP BY course-id;
```

TABLE

course-id	enrolments-count
101	1
102	2
103	3
104	4
105	5

FUNDAMENTALS DATA ANALYTICS EXERCISE 2

Q11 Q1 THE SALARIES TABLE

TABLE SALARIES

employee-id	name	department	salary	bonus
1	Tom	IT	60000	5000
2	Jerry	Jerry HR	55000	4000
3	Spike	Finance	70000	6000
4	Tyler	IT IT	62000	5500
5	Butch	Butch HR	54000	3500

Q11 find total salary and total bonus per department
EXPECTED columns: department, total-salary, total-bonus

A11 SYNTAX

SELECT department

SUM (salary) AS total-salary

SUM (bonus) AS total-bonus

FROM employees

GROUP BY department;

department	total-salary	total-bonus
IT	122000	10500
Finance	70000	6000
HR	109000	7500

DATA ANALYSIS EXERCISE 2

4 THE SALARIES TABLE

Q12 Show department where salary is above 55000.
EXPECTED columns: department, avg-salary

TABLE Salaries

employee-id	name	department	salary	bonus
1	Tom	IT	60 000	5000
2	Jerry	HR	55 000	4000
3	Spike	Finance	70 000	6000
4	Tyke	IT	62 000	5500
5	Butch	HR	54 000	3500

Q12 SYNTAX

```
SELECT department,  
       AVG(salary) AS avg_salary  
FROM employees  
GROUP BY department  
HAVING AVG(salary) > 55000;
```

department	avg - salary
IT	62 000
HR	55 000
Finance	70 000
IT	62 000

DATA ANALYSIS EXERCISE 2

Q13 Select employees whose salary plus bonus is greater than 60000.

EXPECTED columns: employee_id, name, salary, bonus, total compensation.

TABLE Salaries

employee_id	name	department	salary	bonus
1	Tom	IT	60000	5000
2	Jerry	HR	55000	4000
3	Spike	Finance	70000	6000
4	Tyke	IT	62000	5500
5	Butch	HR	54000	3500

A13 SYNTAX

```
SELECT employee_id, name, salary, bonus,
       (salary + bonus) AS total_compensation
FROM employees
WHERE (salary + bonus) > 60000;
```

	employee_id	name	salary	bonus	total_compensation
	1	Tom	60000	5000	65000
<	2	Jerry	55000	4000	59000
	3	Spike	70000	6000	76000
	4	Tyke	62000	5500	67500
<	5	Butch	54000	3500	57500

TABLE

	employee_id	name	salary	bonus	total_compensation
	1	Tom	60000	5000	65000
✓	3	Spike	70000	6000	76000
	4	Tyke	62000	5500	67500

DATA ANALYSIS EXERCISE 2

Q14

Projects TABLE

project-id	Project-name	department	budget
1	AI App	IT	120 000
2	Payroll System	Finance	80 000
3	Dashboard	IT	150 000
4	Website	Marketing	60 000
5	HR Portal	HR	50 000

Q14 Show total and average budget per department. Only include department with average budget above 70 000. EXPECTED COLUMNS: department, total-budget, avg-budget

A14

SYNTAX

SELECT department, SUM (budget) AS total-budget
AVG (budget) AS avg-budget

FROM departments

GROUP BY department;

HAVING AVG (budget) > 70 000

department	total-budget	avg-budget
IT	270 000	195 000
Finance	80 000	80 000

TABLE

department	total-budget	avg-budget
IT	270 000	195 000
Finance	80 000	80 000
Marketing	60 000	60 000
HR	50 000	50 000

1000
1000

DATA ANALYSIS EXERCISE 2

Q15

THE PROJECT TABLE

Project_id	Project-name	department	budget
1	AI APP	IT	120 000
2	Payroll System	Finance	80 000
3	Dashboard	IT	150 000
4	Website	Marketing	60 000
5	HR Portal	HR	50 000

Q15 List all projects with budget between 50 000 and 120 000 excluding the marketing department.
 EXPECTED Column: project-id; project-name; department; budget

Q15 SYNTAX

SELECT * ~~FROM~~

FROM projects

WHERE budget BETWEEN 50 000 AND 120 000
 AND department <> 'Marketing';

TABLE

project_id	project-name	department	budget
2	Payroll System	Finance	80 000

0

0